FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

8 Week: 19 April 2017

All the indicated numbers refer to the "Problems" section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die "Problems" afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P8.5 (a) $v_A = \sqrt{3.00gR}$ (b) 0.098N downward
P8.7 (a) 4.43 m/s (b) 5.0 m
P8.15 (a) 0.791 m/s (b) 0.531 m/s
P8.22 3.74 m/s
P8.23 (a) $\Delta K = -160$ J (b) $\Delta U = 73.5$ J (c) $f_k = 28.8$ N (d) $\mu_k = 0.679$
P8.26 The air resistance acts like a frictional force (a) 24.5 m/s (b) Explanation (convert your answer in (a)) (c) 206 m (d) Explanation
P8.31 5×10^5 N
P8.45 $H = h + \frac{d^2}{4h}$
P8.47 (a) $K = 2 + 24t^2 + 72t^4$ (b) $a = 12t$, $\sum F = 48t$ (c) $P = 48t + 288t^3$ (d) 1250 J
P8.63 $\mu_k = 0.328$
P8.64 $v = 1.24$ m/s
P8.73 Proof that $T_{\text{bottom}} = T_{\text{top}} + 6mg$